

Module-4

Part 1 - Stability Analysis :

- Concepts of Stability
- Necessary conditions for stability
- Routh stability criterion
- Relative stability analysis
- more on the Routh stability Criterion.

Part-2 - Root Locus :

- Introduction to Root Locus Techniques
- The root locus concepts
- Construction of root loci.

Part-3 - Frequency domain analysis & stability :

- Correlation between time & frequency response
- Bode plots
- Experimental determination of transfer function.

Stability Analysis of Control Systems

Transfer Function:

$$TF = \frac{C(s)}{R(s)} = \frac{a_0 s^m + a_1 s^{m-1} + a_2 s^{m-2} + \dots + a_m}{b_0 s^n + b_1 s^{n-1} + b_2 s^{n-2} + \dots + b_n}$$

The numerator & denominator can be factorised to get factorised form of the TF as,

$$TF = \frac{k(s-s_a)(s-s_b)\dots(s-s_m)}{(s-s_1)(s-s_2)\dots(s-s_n)} \quad k \rightarrow \text{system gain factor}$$

Poles of a TF: The value of 's', which make the TF = ∞ after substitution in the denominator of a TF are called poles of that TF. It is indicated by 'x' in s-plane.

eg: 1. $T(s) = \frac{2(s+2)}{s(s+4)}$ $s(s+4) = 0$
 $s=0, s=-4$

If these values are used in dr,
 $TF = \infty$

2. $T(s) = \frac{2(s+2)}{(s+4)^2 (s^2+2s+2)(s+1)}$

$$(s+4)^2 (s^2+2s+2)(s+1) = 0$$

$$s = -4, -4, -1 \pm j, -1$$

simple pole at $s = -1$

complex conjugate poles at $s = -1 \pm j$

repeated pole at $s = -4$

Zeros of a TF: The value of 's', which make the TF = 0 after substituting in the numerator are called zeros of that TF.

eg: $T(s) = \frac{2(s+1)^2 (s+2)(s^2+2s+2)}{s^3(s+4)(s^2+6s+25)}$

simple zero at $s = -2$

repeated zero at $s = -1$

Complex conjugate zeros at $s = -1 \pm j$

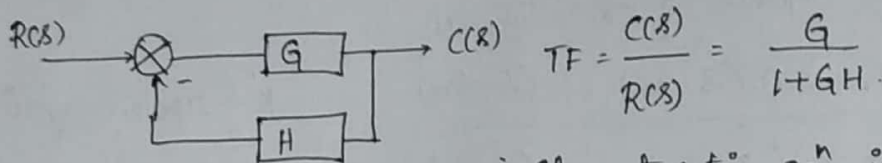
zeros are indicated by 'o' on the s-plane.

Characteristic eqⁿ of a TF:

The eqⁿ obtained by equating denominator to zero, whose roots are the poles of that TF is called characteristic eqⁿ of that system.

$$F(s) = b_0 s^n + b_1 s^{n-1} + b_2 s^{n-2} + \dots + b_n = 0 \rightarrow \text{characteristic eq.}$$

for a -ve feedback closed loop control system,



$\therefore$ Characteristic eqⁿ is $1 + GH = 0$

Order of a TF:

The highest power of 's' present in the characteristic eqⁿ is called Order of a system TF.

Stability of Control System:

Dec 06, Jan 11, July-02

The stability of a linear closed loop system can be determined from the locations of closed loop poles in the s -plane.

$$\text{Eq:- } TF = \frac{C(s)}{R(s)} = \frac{10}{(s+2)(s+4)}$$

to find out, output response for unit step input.

$$P(s) = 1/8$$

$$C(s) = \frac{10}{s(s+2)(s+4)} = \frac{A}{s} + \frac{B}{s+2} + \frac{C}{s+4} = 10 \left\{ \frac{1/8}{s} - \frac{1/4}{s+2} + \frac{1/8}{s+4} \right\}$$

$$(18) = \frac{1.25}{8} - \frac{2.5}{8+2} + \frac{1.25}{8+4}$$

$$C(t) = 1.25 - 2.5e^{-2t} + 1.25e^{-4t}$$

↓
Steady state

↓
Transient
 C_t

$$C(t) = C_{ss} + C_1(t)$$

$$C(t) = 1.25 - 1.5e^{-2t} + 1.25e^{-4t}$$

e^{-2t}, e^{-4t} , as $t \rightarrow \infty$ $C(t) = 0$.

Such systems are called Absolutely stable systems.

Observation: Closed loop poles $s = -2, -4$ & -ve indices of exponential terms are also -2 & -4 .

Thus if closed loop poles are located in left half, exponential indices in o/p are -ve & will vanish as $t \rightarrow \infty$.

eg 2: $\frac{C(s)}{R(s)} = \frac{10}{(s-2)(s+4)}$, $R(s) = \frac{1}{s}$

$$C(s) = \frac{10}{s(s-2)(s+4)} = \left\{ \frac{A}{s} + \frac{B}{s-2} + \frac{C}{s+4} \right\} = \frac{-1.25}{s} + \frac{0.833}{s-2} + \frac{0.416}{s+4}$$

$$C(t) = -1.25 + 0.833e^{2t} + 0.416e^{-4t}$$

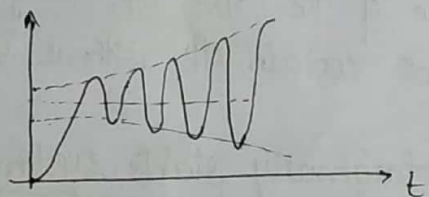
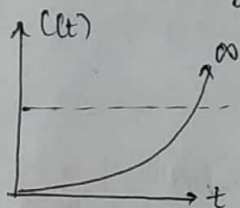
$e^{+2t} \rightarrow$ due to pole located @ right half at $s = 2$

t	C(t)
0	0
1	+4.91
2	+44.23
4	+2481.88
∞	∞

as $t \rightarrow \infty$, due to e^{+2t} , $C(t) \rightarrow$ increases in amplitude.

Such system is said to be unstable.

In such systems o/p is uncontrollable & unbounded one.



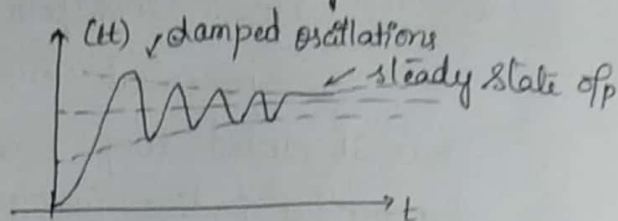
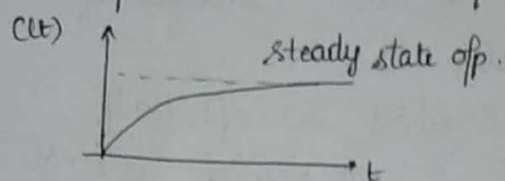
increasing exponentially increasing amplitude oscillations.

For such unstable systems, if i/p = 0, o/p $\neq 0$ & as soon as i/p power is turned on, o/p tends to ∞ .

If no saturation takes place in system & no mechanical stop is provided then system may get damaged & fails.

Conclusion: stability depends on locations of closed loop poles & are the roots of characteristic eqⁿ of the system.

If all the closed loop poles lie in LH of s-plane then o/p response reaches to equilibrium & attains steady state value



Definition of BIBO (Bounded input Bounded output) stability.

Jan 11, July 04
(4m)

Stable system:

A LTI system is said to be stable if,

- 1) When the system is excited by a bounded i/p, o/p is also bounded & controllable.
- 2) In the absence of the i/p, o/p must tend to zero irrespective of the initial conditions.

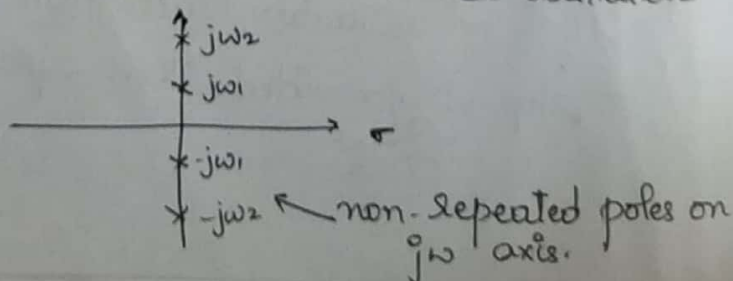
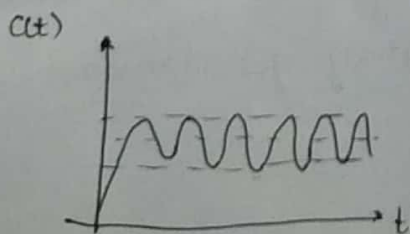
Unstable systems: A linear time invariant system is said to be unstable if,

- 1) For a bounded i/p it produces unbounded o/p
- 2) In absence of the i/p, o/p may not return to zero. It shows certain o/p without i/p.

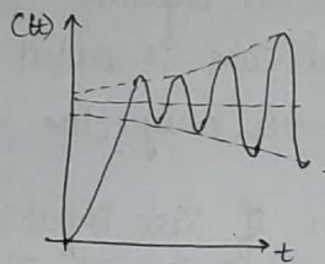
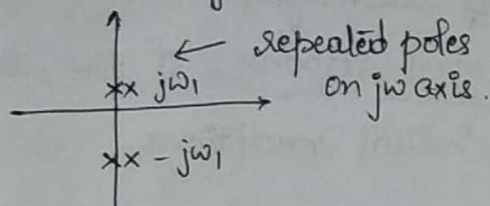
Critically or Marginally stable system:

Jan 11, July 04
(4m)

A LTI system is said to be critically or marginally stable if for a bounded i/p its o/p oscillates with constant frequency & amplitude. Such oscillations of o/p are called undamped oscillation or sustained oscillations



Special case: If there are repeated roots located purely on imaginary axis, system is said to be unstable.



Conditionally Stable system:

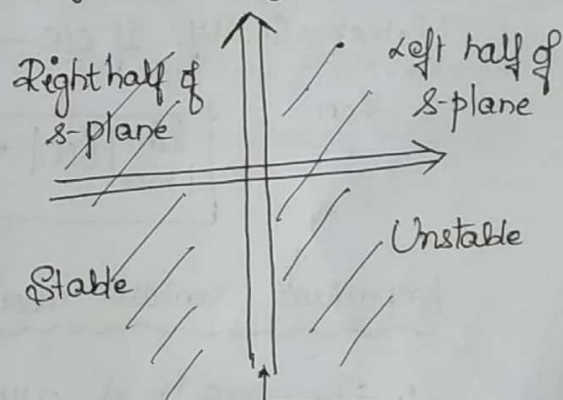
A LTI system is said to be conditionally stable if for a certain condition of a particular parameter of the system, its o/p is bounded one. Otherwise if that condition is violated, o/p becomes unbounded & system becomes unstable. Such system is called Conditionally stable system.

Zero input & Asymptotic Stability.

Zero input stability :

Some systems may get driven by the initial conditions, without any i/p applied to it.

Eg:- A series RC circuit with capacitor initially charged to some voltage. This initial voltage is enough to start the operation of the system. This initial voltage, without any external i/p drives the current till capacitor gets fully discharged. The stability of such a system which is under zero input condition but operated under initial condition is called Zero i/p stability of the system.



Non-repeated roots
→ Marginally stable
Repeated roots
→ Unstable.

Zero input response - The current through RC circuit reduces to zero as capacitor gets fully discharged. The current in such a case is called zero input response of the system, which is only due to the initial conditions.

Definition of zero input stability or Asymptotic stability.

If the zero i/p response of the system subjected to the finite initial conditions, reaches to zero as time t approaches infinity, the system is said to be zero input stable. Otherwise it is called zero i/p unstable.

The zero i/p stability is also called the asymptotic stability.

Mathematically, if $c(t) \rightarrow$ zero i/p response of the system

then

$$\lim_{t \rightarrow \infty} |c(t)| = 0.$$

Important remarks about zero i/p or asymptotic stability:

1. The zero i/p or asymptotic stability depends on the roots of the characteristic eqⁿ i.e. closed loop poles of the system.
2. For asymptotic stability also, all the roots of char eqⁿ must be located in left half of s-plane.
3. If a system is BIBO stable, then it must be zero i/p & asymptotically stable. Thus the system is said to be just stable, unstable or marginally stable for all practical purpose.

Note: non-repeated pair of roots on $j\omega$ axis, $\rightarrow$ system is marginally stable.
Integrator having TF $= 1/s \rightarrow$ root at origin is treated to be stable for all practical purpose as an exception.

Explain the concept of relative stability.

Relative Stability:

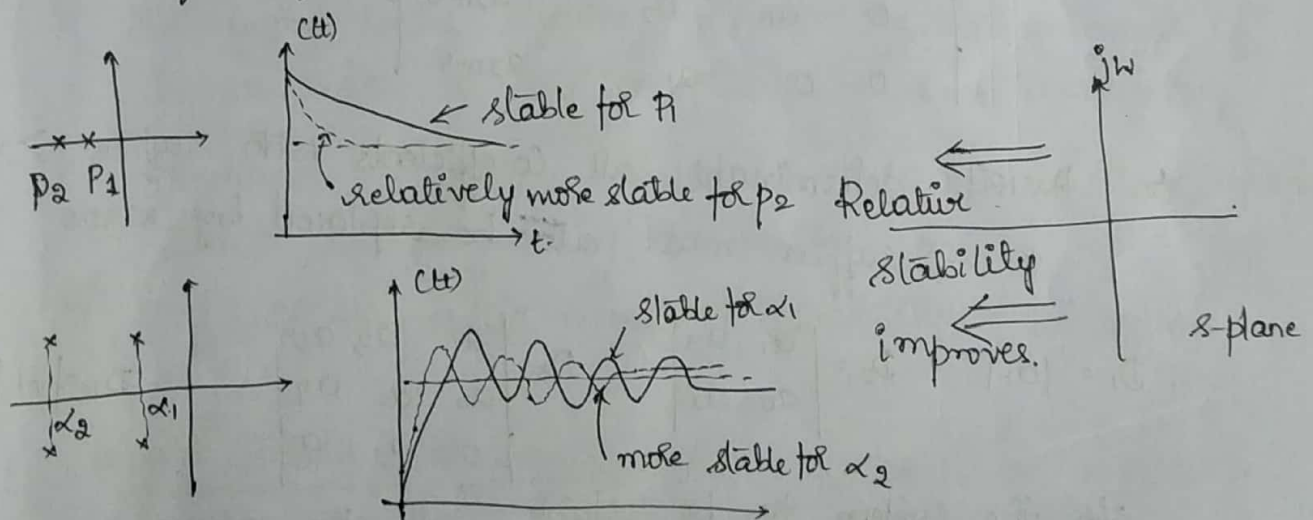
Jan-05, Jan-16 (4m)

The system is said to be relatively more stable or unstable on the basis of settling time. System is said to be relatively more stable if settling time for that system is less than that of the other system.

The settling time of the root or pair of complex conjugate roots is inversely proportional to the real part of the roots, i.e. roots located near $j\omega$ axis have large settling time.

As roots or pair of complex conjugate roots moves away from $j\omega$ axis i.e. towards left half of s -plane, settling time becomes lesser & hence system becomes more & more stable.

Thus relative stability of the system improves, as the closed loop poles move away from the imaginary axis in left half of s -plane.



1. Hurwitz's Criterion:

The sufficient condition for having all the roots of char eqⁿ in the left half of s-plane is given by Hurwitz. It is referred as Hurwitz Criterion.

Statement:

It states that:

"The necessary & sufficient condition to have all roots of characteristic eqⁿ in left half of s-plane is that the sub-determinants D_k , $k=1, 2, \dots, n$ obtained from Hurwitz's determinant 'H' must all be positive."

Method of forming Hurwitz determinant:

$$H = \begin{vmatrix} a_1 & a_3 & a_5 & \dots & a_{2n-1} \\ a_0 & a_2 & a_4 & \dots & a_{2n-2} \\ 0 & a_1 & a_3 & \dots & a_{2n-3} \\ 0 & a_0 & a_2 & \dots & a_{2n-4} \\ 0 & 0 & a_1 & \dots & a_{2n-5} \end{vmatrix}$$

In Hurwitz determinant all Co-efficients with suffices $> n$ & $-ve$ suffices must all be replaced by zeros.

$$D_1 = |a_1| \quad D_2 = \begin{vmatrix} a_1 & a_3 \\ a_0 & a_2 \end{vmatrix} \quad D_3 = \begin{vmatrix} a_1 & a_3 & a_5 \\ a_0 & a_2 & a_4 \\ 0 & a_1 & a_3 \end{vmatrix} \quad \dots \quad D_k = |H|$$

for the system to be stable, all the above determinants must be positive.

Problem:

Determine the stability of the given characteristic eqⁿ by Hurwitz's method.

$$F(s) = s^3 + s^2 + s + 4 = 0$$

Solⁿ: $a_0 = 1 \quad a_1 = 1 \quad a_2 = 1 \quad a_3 = 4 \quad n = 3$
 $a_4 = 2 \quad a_5 = 0$

$$H = \begin{vmatrix} a_1 & a_3 & a_5 \\ a_0 & a_2 & a_4 \\ 0 & a_1 & a_3 \end{vmatrix} = \begin{vmatrix} 1 & 4 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 4 \end{vmatrix}$$

$$D_1 = |1| = 1$$

$$D_2 = \begin{vmatrix} 1 & 4 \\ 1 & 1 \end{vmatrix} = -3$$

$$D_3 = \begin{vmatrix} 1 & 4 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 4 \end{vmatrix} = 4 - 16 = -12$$

As D_2 & $D_3 = -ve$, system is unstable.

Disadvantages of Hurwitz's method.

- 1> For higher order systems, to solve the determinants of higher order is very complicated & time consuming.
- 2> Number of roots located in RHS for unstable system cannot be judged by this method.
- 3> Difficult to predict marginal stability of the system.

Routh Stability Criterion:

July 04, 09, 11, 12, 14, 17.

Feb 10, Jan 05, 11, 12, 15, 18, 19

To overcome limitations of Hurwitz's method, Scientist Routh suggested a new method called Routh's method or Routh-Hurwitz method or Routh's array Method.

Routh's array: Routh suggested a method of tabulating the Co-efficients of characteristic equation, which gives an array called Routh's array.

Consider the general characteristic eqⁿ as

$$F(s) = a_0 s^n + a_1 s^{n-1} + a_2 s^{n-2} + \dots + a_n = 0$$

Let $n=5$

$$F(s) = a_0 s^5 + a_1 s^4 + a_2 s^3 + a_3 s^2 + a_4 s + a_5 = 0.$$

$$s^5 \quad a_0 \quad a_2 \quad a_4$$

$$s^4 \quad a_1 \quad a_3 \quad a_5$$

$$s^3 \quad b_1 \quad b_2$$

$$s^2 \quad c_1 \quad c_2$$

$$s^1 \quad d_1$$

$$s^0 \quad e_1$$

$$b_1 = \frac{a_1 a_2 - a_0 a_3}{a_1}$$

$$b_2 = \frac{a_1 a_4 - a_0 a_5}{a_1}$$

$$c_1 = \frac{b_1 a_3 - a_1 b_2}{c_1}$$

$$c_2 = \frac{b_1 a_5 - a_1(0)}{c_1}$$

What are the necessary & sufficient cond^s for the sys to be stable as per R-H criteria
Routh's Criterion

The necessary & sufficient condition

$$d_1 = \frac{c_1 b_2 - b_1 c_2}{c_1}$$

Jan 15
(4m)

The necessary & sufficient condition for system to be stable is "All the terms in the first column of Routh's array must have same sign. There should not be any sign change in the first column of Routh's array."

If there are any sign changes existing then,

a) System is unstable

- b) The number of sign changes equals the no. of roots in R.H.S.

Advantages of Routh's criterion

July 16 (4m)

Advantages of Routh's array method are:

- 1) Stability of the system can be judged without actually solving the char eqⁿ.
- 2) No evaluation of determinants, which saves calculation time.
- 3) For unstable system it gives no. of roots of char eqⁿ having +ve real part.
- 4) Relative stability of the system can be easily found.
- 5) Critical value of system gain can be determined hence frequency of sustained oscillations can be found.
- 6) It helps in finding out range of values of K for system stability.
- 7) It helps in finding out intersection points of root locus with imaginary axis.

Limitations of Routh's criterion

July -09, 11, 12, 14, 15
16, 18 Jan -05, 12
(3m)

- 1) It is valid only for real coefficients of the characteristic eqⁿ.
- 2) It does not provide exact locations of the closed loop poles in left or right half of s -plane.
- 3) It does not suggest methods of stabilising an unstable system.
- 4) Applicable only to linear systems.

Applications of Routh's Criterion:

April-96 Aug-01, 03, 05, 08
Feb-03, March-02, June-12, 13
July-06, 07, 12, 13, 14, 16, 17, 18, 19
Jan-11, 14, 16, 18

1. Relative Stability Analysis.

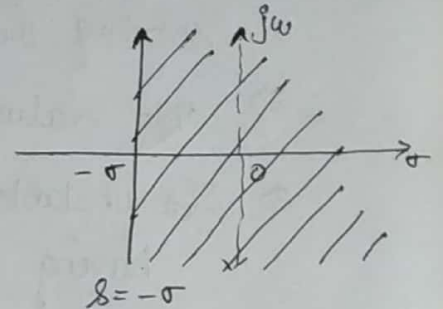
To find relative stability of system about a line $s = -\sigma$ is to find how many roots are located in right half of this line $s = -\sigma$, the Routh's method is used effectively.

To find, shift the axis of s -plane & then apply Routh array & substitute $s = s' - \sigma$ ($\sigma \rightarrow \text{constant}$) in char eqⁿ.

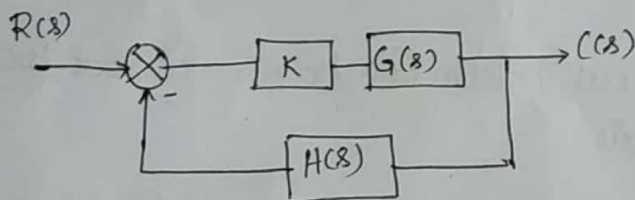
Write polynomial in terms of s' .

Complete array from this new eqⁿ.

The no. of sign changes in 1st column = no. of roots located in right of $s = -\sigma$.



2. Determining Range of values of k



In an amplifier of variable gain k is introduced.

$$\frac{C(s)}{R(s)} = \frac{k G(s)}{1 + k G(s) H(s)}$$

$$\therefore \text{char eq}^n \quad \boxed{F(s) = 1 + k G(s) H(s) = 0.}$$

$\therefore$ the location of roots are dependent on the proper selection of value of ' k '.

Routh array can be constructed in terms of k & then the range of values of k can be obtained in such a way that it will not produce any sign change in the

1st column of the Routh's array. Hence it is possible to obtain the range of values of k for absolute stability of the system using Routh's criterion. Such a system where stability depends on the condition of parameter ' k ' is called conditionally stable system.

Special cases of Routh's Criterion

March-02, Aug-03, 6

Feb-09 Dec-11

Jan-09, 11, 14, 15, 18, 19 June 13

July 17, 18, 19

Special case 1

First element of any of the rows of Routh's array is zero & the remaining are non-zero element.

Ex:- $s^5 + 2s^4 + 3s^3 + 6s^2 + 2s + 1 = 0$

$$s^5 \quad 1 \quad 3 \quad 2$$

$$s^4 \quad 2 \quad 6 \quad 1$$

$$s^3 \quad 0 \quad 1.5 \quad 0 \quad \text{Special case I}$$

$$s^2 \quad \infty \quad \dots$$

$$s^1$$

$$s^0$$

Solⁿ: 1st method

Substitute ' ϵ ' - small +ve no. in place of zero.

Complete array with ϵ , then examine the sign change by taking $\lim_{\epsilon \rightarrow 0}$.

$$s^5 \quad 1 \quad 3 \quad 2$$

$$s^4 \quad 2 \quad 6 \quad 1$$

$$s^3 \quad \epsilon \quad 1.5$$

$$s^2 \quad \frac{6\epsilon - 3}{\epsilon} \quad 1$$

$$s^1 \quad \frac{1.5(6\epsilon - 3) - \epsilon}{\epsilon} \quad \frac{(6\epsilon - 3)/\epsilon}{\epsilon}$$

$$s^0 \quad 1$$

To examine sign changes,

$$\lim_{e \rightarrow 0} \left(\frac{6e-3}{e} \right) = \lim_{e \rightarrow 0} \left(6 - \frac{3}{e} \right) = 6 - \lim_{e \rightarrow 0} \left(\frac{3}{e} \right) = 6 - \infty = -\infty$$

$$\lim_{e \rightarrow 0} \frac{1.5(6e-3) - e^2}{(6e-3)} = \lim_{e \rightarrow 0} \frac{9e - 4.5 - e^2}{6e-3} = \frac{-4.5}{-3} = +1.5$$

sign $\rightarrow$ -ve

sign $\rightarrow$ +ve

$\therefore$ Routh's array is,

s^5	1	3	2
s^4	2	6	1
s^3	$+e$	1.5	0
s^2	$\downarrow$ $-\infty$	1	0
s^1	$\downarrow$ $+1.5$	0	0
s^0	1	0	0

As there are two sign changes,
system is unstable.

2nd method:

Replace $s \rightarrow 1/z$ in original eqⁿ, then taking LCM of x^4 with ~~lowest~~ highest power of z , rearrange char eqⁿ in descending powers of z , complete the array & examine the stability with this array.

$$F(s) = s^5 + 2s^4 + 3s^3 + 6s^2 + 2s + 1 = 0.$$

$$s \rightarrow 1/z \quad \frac{1}{z^5} + \frac{2}{z^4} + \frac{3}{z^3} + \frac{6}{z^2} + \frac{2}{z} + 1 = 0$$

x^4 by z^5

$$z + 2z + 3z^2 + 6z^3 + 2z^4 + z^5 = 0$$

$$\therefore z^5 + 2z^4 + 6z^3 + 3z^2 + 2z + 1 = 0.$$

z^5	1	6	2
z^4	2	3	1
z^3	4.5	1.5	0
z^2	2.33	1	0
z^1	-0.429	0	
z^0	1		

As there are two sign changes,
system is unstable.

Special case 2

eg:- $F(s) = s^6 + 3s^5 + 4s^4 + 6s^3 + 5s^2 + 3s + 2 = 0$

$$s^6 \quad 1 \quad 4 \quad 5 \quad 2$$

$$A(s) = 2s^4 + 4s^2 + 2 = 0$$

$$s^5 \quad 3 \quad 6 \quad 3 \quad 0$$

$$A(s) = s^4 + 2s^2 + 1 = 0$$

$$s^4 \quad 2 \quad 4 \quad 2 \quad 0$$

$$\frac{dA(s)}{ds} = 4s^3 + 4s$$

$$s^3 \quad 0(4) \quad 0(4) \quad 0 \quad 0 \leftarrow \text{special case 2}$$

$$s^2 \quad 2 \quad 2$$

$$A'(s) = 2s^2 + 2 = 0$$

$$s^1 \quad 0(4) \quad 0 \leftarrow \text{special case 2}$$

$$\frac{dA'(s)}{ds} = 4s = 0$$

$$s^0 \quad 2$$

→ No sign change, hence no root is located in RHS of s-plane.

→ As row of zero occurs, system may be marginally or unstable.

→ ∴ Examine roots of 1st Auxiliary eqⁿ.

$$A(s) = 2s^4 + 2s^2 + 1 = 0$$

$$y = s^2 \quad y^2 + 2y + 1 = 0$$

$$a = 1$$

$$b = 2$$

$$c = 1$$

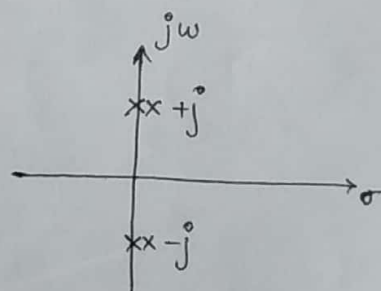
$$y = \frac{-2 \pm \sqrt{4 - 4}}{2} = -1, -1$$

$$s_{1,2}^2 = -1, -1$$

$$s_{3,4}^2 = -1, -1$$

$$s_{1,2} = \pm j, \pm j$$

$$s_{3,4} = \pm j, \pm j$$



→ Roots of $A'(s) = 0$ are the roots of $A(s) = 0$. So do not solve 2nd auxiliary eqⁿ.

→ As there are repeated roots on imaginary axis, system is unstable.

Marginal k (k_{mar})

Marginal value of ' k ' is value of ' k ' which makes any row of Routh array as row of zeros except s^0 is called Marginal value of k (k_{mar}).

Frequency of sustained oscillations

To obtain the frequency of oscillations, solve the auxiliary eqⁿ $A(s)=0$ for $k=k_{mar}$.

The imaginary roots of $A(s)=0$, obtained for k_{mar} indicates the frequency of sustained oscillations.

Problems

1. A system has the char eqⁿ $s^3 + 9s^2 + 26s + 24 = 0$ using R-H criterion judge for the stability of the system.

Solⁿ:

s^3	1	26
s^2	9	24
s^1	$70/3 (b_1)$	0
s^0	24 (a ₁)	

$$b_1 = \frac{9 \times 26 - 1 \times 24}{9} = 70/3$$

$$c_1 = \frac{70/3 \times 24 - 9 \times 0}{70/3} = 24$$

Conclusion: Since there are no sign changes in the 1st column of the Routh's array indicating that all the roots lie in the left half of the s -plane & hence the system is stable.

2. $s^5 + 3s^4 + 7s^3 + 20s^2 + 6s + 15 = 0$

s^5	1	7	6
s^4	3	20	15
s^3	$(b_1) 1/3$	$(b_2) 1$	
s^2	$(c_1) 11$	$(c_2) 15$	
s^1	$(d_1) 6/11$		
s^0	$(e_1) 15$		

$$b_1 = \frac{3 \times 7 - 1 \times 20}{3} = 1/3$$

$$b_2 = \frac{3 \times 6 - 1 \times 15}{3} = \frac{18 - 15}{3} = 1$$

$$c_1 = \frac{1/3 \times 20 - 3 \times 1}{1/3} = \frac{20/3 - 3}{1/3} = 11$$

$$d_1 = \frac{6/11 \times 15 - 0}{6/11} = 15$$

$$c_2 = \frac{1/3 \times 15 - 0}{1/3} = 15$$

Conclusion: Since there are no sign change in the 1st column

of the Routh array indicates that all the roots lie in the left half of the s -plane & hence it is stable.

3. $s^4 + 2s^3 + 3s^2 + 8s + 2 = 0$.

Solⁿ:

s^4	1	3	2
s^3	+ 2	8	
s^2	{	→	2
s^1			
s^0	2		

Conclusion: Since there are 2 sign changes (+ to -ve & -ve to +) in the 1st column of Routh's array, the system is unstable. As there are 2 sign changes 2 roots out of 4 roots lie in R.H.S & remaining 2 root lies in L.H.S.

Stability Analysis:

Preeti T.S.
Lecturer
JVI, Bichali

$$TF = \frac{P(s)}{Q(s)} = \frac{a_0 s^m + a_1 s^{m-1} + a_2 s^{m-2} + \dots + a_m}{b_0 s^n + b_1 s^{n-1} + b_2 s^{n-2} + \dots + b_n}$$

The numerator & denominator can be factorised to get the factorised form of the T.F as,

$$T.F = \frac{k(s-s_a)(s-s_b) \dots (s-s_m)}{(s-s_1)(s-s_2) \dots (s-s_n)}$$

$k \rightarrow$ system gain factor.

Poles of a T.F: The values of 's', which make the T.F infinite after substitution in the denominator of a T.F are called poles of that T.F.

Eg: $T(s) = \frac{2(s+2)}{s(s+4)}$

To obtain poles,

$s(s+4) = 0$ poles are real.

$s=0, s=-4$. If these values are used in the denominator, the value of T.F = ∞ .

Eg: $T(s) = \frac{2(s+2)}{(s+4)^2 (s^2+2s+2)(s+1)}$

$$(s+4)^2 (s^2+2s+2)(s+1) = 0$$

$$s = -4, -4, -1 \pm j, -1$$

$T(s)$ has simple pole at $s = -1$, complex conjugate poles at $s = -1 \pm j$, repeated pole at $s = -4$.
Poles are indicated by 'x' (cross) in s-plane.

Zeros of a T.F: The values of s which make the T.F zero after substituting in the numerator are called zeros of that T.F.

$$\text{eg: } T(s) = \frac{2(s+1)^2(s+2)(s^2+2s+2)}{s^3(s+4)(s^2+6s+25)}$$

$$2(s+1)^2(s+2)(s^2+2s+2)=0$$

Simple zero at $s=-2$, repeated zero at $s=-1$ (twice)

Complex Conjugate zeros at $s=-1 \pm j$

Zeros are indicated by small $\circ$ or $\circ$ zero in the s -plane.

Characteristics eqⁿ of a T.F:

The eqⁿ obtained by equating denominator of a T.F to zero, whose roots are the poles of that T.F is called Char. eqⁿ of that system.

$F(s) = b_0 s^n + b_1 s^{n-1} + b_2 s^{n-2} + \dots + b_n = 0$ is called the char eqⁿ.

Order of a T.F

The highest power of s present in the Char eqⁿ is in the denominator polynomial of a closed loop T.F of a system is called order of a system.

Pole-zero plot: $\rightarrow$ Plot obtained by locating all poles & zeros of a T.F in s -plane is called pole-zero plot of a system.

Stability of Control Systems:

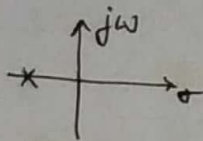
The stability of a linear closed loop system can be determined from the locations of closed loop poles in the s -plane.

Nature of closed loop poles

Location

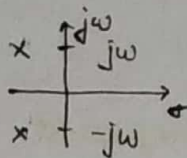
Stability

1) Real -ve



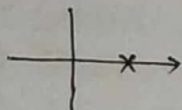
Stable

2) Complex conjugate with -ve real pole



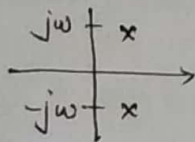
Stable

3) Real +ve



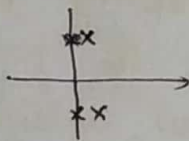
Unstable

4) Complex conjugate +ve real pole



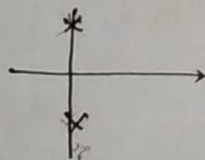
Unstable

5) Complex conjugate repeated poles on imaginary axis



Unstable

6) Non-repeated poles & imaginary poles.



Marginally /
Partially /
Critically
stable

1) Investigate the stability of a closed loop system whose characteristic is given by, $s^5 + s^4 + 2s^3 + 2s^2 + 3s + 5 = 0$.

Solⁿ:

s^5	1	2	3
s^4	1	2	5
s^3	0	-2	0
s^2			
s^1			
s^0			

As we encountered zero in 1st element s^3 & other elements are not zero.

This difficulty can overcome by replacing s with $(\frac{1}{z})$.

$$\left(\frac{1}{z}\right)^5 + \left(\frac{1}{z}\right)^4 + 2\left(\frac{1}{z}\right)^3 + 2\left(\frac{1}{z}\right)^2 + 3\left(\frac{1}{z}\right) + 5 = 0$$

$$\frac{1}{z^5} + \frac{1}{z^4} + 2 \cdot \frac{1}{z^3} + 2 \cdot \frac{1}{z^2} + 3 \cdot \frac{1}{z} + 5 = 0$$

$\times z^5$ by z^5 & rearranging,

$$5z^5 + 3z^4 + 2z^3 + 2z^2 + z + 1 = 0$$

$$z^5 \quad 5 \quad 2 \quad 1$$

$$z^4 \quad 3 \quad 2 \quad 1$$

$$z^3 \quad -4/3 \quad -2/3 \quad 0$$

$$z^2 \quad 1/2 \quad 1$$

$$z^1 \quad 2$$

$$z^0 \quad 1$$

Since there is ~~no~~ 2 sign change, 2 roots lie on left right half of the s -plane.
 $\therefore$ Sys is unstable.

2) $s^6 + 2s^5 + 5s^4 + 8s^3 + 8s^2 + 8s + 4 = 0$

Solⁿ:

s^6	1	5	8	4
s^5	2	8	8	0
s^4				
s^3	1	4	4	
s^2	0	0	0	
s^1				
s^0				

Aug-03 Dec11, June13
(6m)

$$A(s) = s^4 + 4s^2 + 4$$

$$\frac{dA(s)}{ds} = 4s^3 + 8s$$

s^6	1	5	8	4
s^5	2	8	8	
s^4	1	4	4	
s^3	4	8		
s^2	2	4		
s^1	4	0		
s^0	4			

$$A(s) = 2s^2 + 4$$

$$\frac{dA(s)}{ds} = 4s$$

$$s^4 + 4s^2 + 4 = 0$$

$$2s^2 + 4 = 0$$

$$(s^2)^2 + 2s^2 + 2s^2 + 4 = 0$$

$$s^2 + 2 = 0$$

$$s^2(s^2 + 2) + 2(s^2 + 2) = 0$$

$$s^2 = -2$$

$$(s^2 + 2) = 0 \quad (s^2 + 2) = 0$$

$$s = \pm j\sqrt{2}$$

$$s^2 = -2 \quad s^2 = -2$$

$$s = \pm j\sqrt{2} \quad s = \pm j\sqrt{2}$$

Since there is no sign changes no root lies in the right half of the s -plane, but $\because$ of zero element it may be marginally stable. To examine this we need to have the roots of the auxiliary eq.

Since we have roots like $s = \pm j\sqrt{2}$ & $s = \pm j\sqrt{2}$ is repeated roots lies on the imaginary axis & as a result system is unstable.

3. $s^6 + 2s^5 + 8s^4 + 12s^3 + 20s^2 + 16s + 16 = 0$

Solⁿ:

s^6	1	8	20	16
s^5	2	12	16	0
s^4	2	12	16	0
s^3	0	0		
s^2				
s^1				
s^0				

$$A(s) = 2s^4 + 12s^2 + 16$$

$$\frac{dA(s)}{ds} = 8s^3 + 24s$$

s^6	1	8	20	16
s^5	2	12	16	0
s^4	2	12	16	0
s^3	8	24		
s^2	6	16		
s^1	2.67			
s^0	16			

Since the roots are non-repeated & as it lies on imaginary axis, the system is marginally stable system.

Since no sign change the system may be stable but $\therefore$ of existence of zero elements row (s^3 row) it may be marginally stable or unstable.

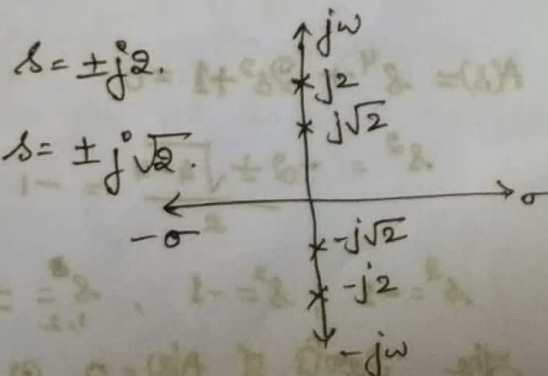
$$2s^4 + 12s^2 + 16 = 0$$

$$s^4 + 6s^2 + 8 = 0$$

$$s^4 + 14s^2 + 2s^2 + 8 = 0$$

$$s^2(s^2 + 4) + 2(s^2 + 4) = 0$$

$$(s^2 + 4) = 0 \quad (s^2 + 2) = 0$$



repeated
Problem
(8.8)

For a system with characteristic eqⁿ $F(s) = s^6 + 8s^5 + 4s^4 + 6s^3 + 5s^2 + 3s + 2 = 0$,
examine stability.

July 19
(8m)

Solⁿ:

$$\begin{array}{l} s^6 \quad 1 \quad 4 \quad 5 \quad 2 \\ s^5 \quad 3 \quad 6 \quad 3 \quad 0 \\ s^4 \quad 2 \quad 4 \quad 2 \quad 0 \\ s^3 \quad \boxed{0 \quad 0 \quad 0 \quad 0} \end{array}$$

Row of zeros.

$$\begin{array}{l} s^6 \quad 1 \quad 4 \quad 5 \quad 2 \\ s^5 \quad 3 \quad 6 \quad 3 \quad 0 \\ s^4 \quad 2 \quad 4 \quad 2 \quad 0 \\ s^3 \quad 4 \quad 4 \quad 0 \quad 0 \\ s^2 \quad 2 \quad 2 \quad 0 \quad 0 \\ s^1 \quad 0 \quad 0 \quad 0 \quad 0 \end{array}$$

Row of zeros again

$$\begin{array}{l} s^6 \quad 1 \quad 4 \quad 5 \quad 2 \\ s^5 \quad 3 \quad 6 \quad 3 \quad 0 \\ s^4 \quad 2 \quad 4 \quad 2 \quad 0 \\ s^3 \quad 4 \quad 4 \quad 0 \quad 0 \\ s^2 \quad 2 \quad 2 \quad 0 \quad 0 \\ s^1 \quad 2 \quad 0 \quad 0 \quad 0 \\ s^0 \quad 2 \end{array}$$

$$A(s) = 2s^4 + 4s^2 + 2 = 0$$

$$s^4 + 2s^2 + 1 = 0$$

$$\frac{dA(s)}{ds} = 4s^3 + 4s$$

$$A'(s) = 2s^2 + 2 = 0$$

$$A'(s) = s^2 + 1 = 0$$

$$\frac{dA'(s)}{ds} = 2s$$

No sign change, hence no root is located in RHS of s-plane

As row of zeros occur, system may be marginally/unstable.

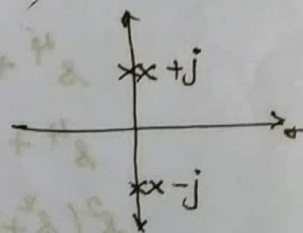
To examine, find the roots of 1st auxiliary eqⁿ,

$$A(s) = s^4 + 2s^2 + 1 = 0 \quad y^2 + 2y + 1 = 0$$

$$s^2 = \frac{-2 \pm \sqrt{4-4}}{2} = -1$$

$$y = \frac{-2 \pm \sqrt{4-4}}{2} = -1$$

$$s^2 = -1, s^2 = -1, s_{1,2}^0 = \pm j, s_{3,4}^0 = \pm j$$



The roots of $A(s) = 0$ as $A(s) = 0$ So don't solve 2nd Auxiliary eqⁿ. $\therefore$ As there are repeated roots on imaginary axis, system is unstable.

5. ^{repeated problem} Check the stability of the given characteristic eqⁿ using Routh's method $s^6 + 2s^5 + 8s^4 + 12s^3 + 20s^2 + 16s + 16 = 0$.

Solⁿ:

s^6	1	8	20	16
s^5	2	12	16	0
s^4	2	12	16	0
s^3	0	0	0	0

s^6	1	8	20	16
s^5	2	12	16	0
s^4	2	12	16	0
s^3	8	24	0	0
s^2	6	16	0	0
s^1	2.67	0	0	0
s^0	16	0	0	0

$$A(s) = 2s^4 + 12s^2 + 16 = 0$$

$$\frac{dA(s)}{ds} = 8s^3 + 24s = 0$$

No sign change, so system may be stable. But as there is row of zero, system will be 1) marginally stable or 2) unstable.

To examine solve $A(s) = 0$

$$2s^4 + 12s^2 + 16 = 0$$

$$s^4 + 6s^2 + 8 = 0$$

$$s^2 = y$$

$$y^2 + 6y + 8 = 0$$

$$y = \frac{-6 \pm \sqrt{36 - 32}}{2}$$

$$\therefore s^2 = -2 \quad s^2 = -4$$

$$= -3 \pm 1 = -2, -4$$

$$s = \pm j\sqrt{2} \quad s = \pm j2$$

Non-repeated roots on imaginary axis. $\therefore$ Sys is marginally stable.

6.

A feedback control system has a characteristic eqⁿ

$$s^6 + 2s^5 + 9s^4 + 16s^3 + 24s^2 + 32s + 16 = 0. \text{ How many poles}$$

are 1) in LHS 2) on the jw axis, & on the RHS.

Jan-19
(8m)

Solⁿ: The Routh's array is,

$$s^6 \quad 1 \quad 9 \quad 24 \quad 16$$

$$s^5 \quad 2 \quad 16 \quad 32 \quad 0$$

$$s^4 \quad 1 \quad 8 \quad 16 \quad 0$$

$$s^3 \quad \boxed{0 \quad 0 \quad 0 \quad 0}$$

$$s^3 \quad 4 \quad 16 \quad 0$$

$$s^2 \quad 4 \quad 16 \quad 0$$

$$s^1 \quad \boxed{0 \quad 0}$$

$$s^1 \quad 8 \quad 0$$

$$s^0 \quad 16 \quad 0$$

$$A(s) = s^4 + 8s^2 + 16 = 0$$

$$\frac{dA(s)}{ds} = 4s^3 + 16s$$

$$A'(s) = 4s^2 + 16$$

$$\frac{dA'(s)}{ds} = 8s$$

As no sign change in 1st Column of array, no root in RHS of s-plane, solving,

$$A(s) = s^4 + 8s^2 + 16 = 0$$

$$\text{Let } y = s^2$$

$$y^2 + 8y + 16 = 0$$

$$y^2 + 4y + 4y + 16 = 0$$

$$y(y+4) + 4(y+4) = 0$$

$$(y+4) = 0 \quad y+4 = 0$$

$$y = -4, \quad y = -4$$

$$s_{1,2}^2 = -4 \quad s_{3,4}^2 = -4$$

$$\boxed{s_{1,2} = \pm j2 \quad s_{3,4} = \pm j2}$$

$\therefore$ Out of 6 roots, 4 roots lie on imaginary axis & remaining 2 roots on LHS of s-plane.

$\therefore$ System is marginally stable.

$$\therefore \begin{cases} \text{Poles in LHS} = 2 \\ \text{Poles in RHS} = 0 \\ \text{Poles on } j\omega \text{ axis} = 4 \end{cases}$$

7. The characteristic eqⁿ for certain feedback control system is given by, $s^4 + 4s^3 + 13s^2 + 36s + K = 0$, determine the range of values of K for which the system is stable.

Solⁿ:

s^4	1	13	K
s^3	4	36	
s^2	4	K	
s^1	$\frac{144 - 4K}{4}$	0	
s^0	K		

Since the system is stable for $K > 0$

$$\frac{144 - 4K}{4} > 0$$

$$144 - 4K > 0$$

$$144 > 4K$$

$$K < \frac{144}{4} \Rightarrow K < 36$$

$\therefore$ The range of K is $0 < K < 36$

Marginal K

The value of ' K ' which makes any row of Routh array as row of zeroes is called Marginal value of K .

To find frequency

→ find out roots of Auxiliary eqⁿ $A(s) = 0$ for $K = K_{mar}$.

→ The magnitude of imaginary roots of $A(s) = 0$ obtained for marginal value of K_{mar} indicates the frequency of sustained oscillations, which system is going to produce.

8. ✓ For unity feedback system,

(8.b) $G(s) = \frac{k}{s(1+0.4s)(1+0.25s)}$; find range of values of k ,

marginal value of k & frequency of sustained oscillations.

Solⁿ: Char. eqⁿ, $1 + G(s) \cdot H(s) = 0$ & $H(s) = 1$

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Jan-19
(8m)

$$1 + GH(s) = 0$$

$$1 + \frac{k}{s(1+0.4s)(1+0.25s)} = 0$$

$$s(1+0.4s)(1+0.25s)$$

$$s[1 + 0.65s + 0.1s^2] + k = 0$$

$$\therefore 0.1s^3 + 0.65s^2 + s + k = 0$$

$$s^3 \quad 0.1 \quad 1$$

$$s^2 \quad 0.65 \quad k$$

$$s^1 \quad \frac{0.65 - 0.1k}{0.65} \quad 0$$

$$s^0 \quad k$$

from, s^0 , $k > 0$

from s^1 , $\frac{0.65 - 0.1k}{0.65} > 0$

$$0.65 - 0.1k > 0$$

$$0.65 > 0.1k$$

$$k < 6.5$$

$\therefore$ Range of values of k ,

$$\boxed{0 < k < 6.5}$$

The marginal value of ' k ' is a value which makes any row other than s^0 as row of zeroes.

$$0.65 - 0.1k_{\text{mar}} = 0$$

$$\boxed{k_{\text{mar}} = 6.5}$$

To find frequency, find out roots of auxiliary eq at marginal value of 'k'.

$$A(s) = 0.65s^2 + k = 0$$

$$0.65s^2 + 6.5 = 0$$

$$s^2 = -10$$

$$s = \pm j 3.162$$

Comparing with $s = \pm j\omega$

$\omega = \text{freq of oscillations}$

$$\therefore \boxed{\omega = 3.162 \text{ rad/sec}}$$

9. ✓ A unity feedback system is characterised by an open loop transfer funⁿ $G(s) = \frac{k(s+13)}{s(s+3)(s+7)}$ using R-H criterion. 1) Calculate the range of values of k for the system to be stable, 2) has its closed loop pole more -ve than -1.

Soln: The char eqⁿ is given by,

$$1 + G(s)H(s) = 0 \quad H(s) = 1$$

$$1 + \frac{k(s+13)}{s(s+3)(s+7)} = 0$$

$$s(s+3)(s+7) + k(s+13) = 0$$

$$s^3 + 10s^2 + (21+k)s + 13k = 0$$

$$s^3 \quad 1 \quad (21+k)$$

$$s^2 \quad 10 \quad 13k$$

$$s^1 \quad 10(21+k) - 13k$$

$$s^0 \quad 13k$$

March 02
Aug-01 July-06
July-09, 16
Jan-16 (8m)

Since the sys is stable,

I

1) s^0 , $13k > 0$ or $k > 0$

2) s^1 , $\underline{210 - 3k} > 0$

$\therefore k < 70$

$\therefore \boxed{0 < k < 70} \rightarrow$ Range of k for stability.

II, For its closed loop poles more negative than -1,

replace $s = s' - 1$ in original char eq,

$$(s'-1)^3 + 10(s'-1)^2 + (s'-1)(k+2) + 13k = 0$$

$$(s')^3 - 3(s')^2 + 3s' + 10(s')^2 + 20s' + 10 + ks' + 21s' - 21 + 13k = 0$$

$$(s')^3 + 7(s')^2 + s'(4+k) + (13k-12) = 0$$

Routh's Array

$(s')^3$	1	$(4+k)$
$(s')^2$	7	$12k-12$
$(s')^1$	$\frac{28+7k-12k+12}{7}$	0
$(s')^0$	$12k-12$	

From $(s')^0$, $12k-12 > 0$

$k > 1$

From $(s')^1$, $-5k+40 > 0$

$40 > 5k$ i.e.

$k < 8$

$\therefore \boxed{1 < k < 8} \rightarrow$ This is the range of k to get closed loop pole more $\rightarrow$ than -1.

(8.7)

10. For system $s^4 + 22s^3 + 10s^2 + s + k = 0$, find k_{mar} & ω at k_{mar} .

Soln:

s^4	1	10	k
s^3	22	1	0
s^2	9.95	k	0
s^1	$\frac{9.95 - 22k}{9.95}$		
s^0	k		

July 15
(6m)

Marginal value of k which makes row of s^1 as row of zeros.

$$\frac{9.95 - 22k_{\text{mar}}}{9.95} = 0$$

$$9.95 - 22k_{\text{mar}} = 0$$

$$9.95 = 22k_{\text{mar}}$$

$$k_{\text{mar}} = \frac{9.95}{22} = 0.4524$$

$$A(s) = 9.95s^2 + k = 0$$

$$9.95s^2 + 0.4524 = 0$$

$$s^2 = -0.04546$$

$$s = \pm j0.2132$$

WKT, $s = \pm j\omega$

$$\therefore \omega = 0.2132 \text{ rad/sec}$$

11. Find the range of values of k so that system with the following characteristics eqⁿ will be stable.

$$s(s^2 + s + 1)(s + 4) + k = 0$$

Solⁿ: $s(s^2 + s + 1)(s + 4) + K = 0$

$$s^4 + 5s^3 + 5s^2 + 4s + K = 0$$

$$s^4 \quad 1 \quad 5 \quad K$$

$$s^3 \quad 5 \quad 4$$

$$s^2 \quad 21/5 \quad K$$

$$s^1 \quad \frac{84 - 25K}{21}$$

$$s^0 \quad K$$

Since the system is stable,

$$K > 0$$

$$\frac{84 - 25K}{21} > 0$$

$$84 - 25K > 0$$

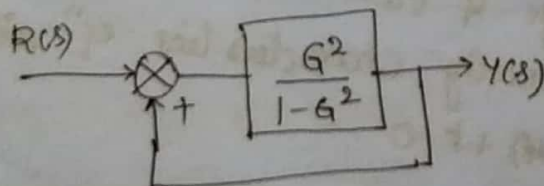
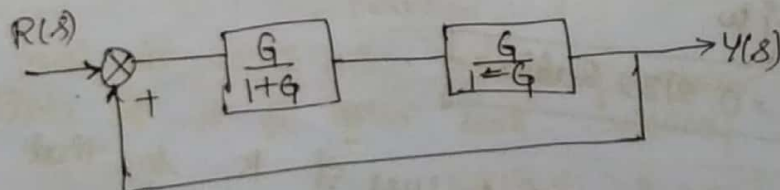
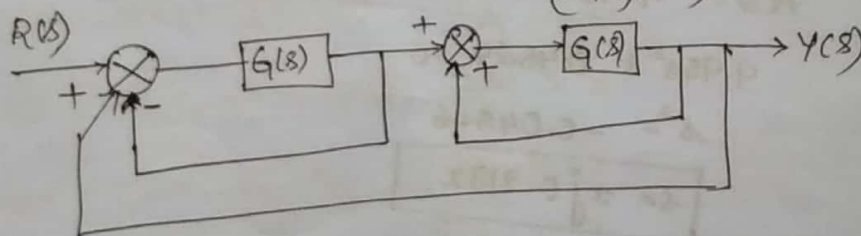
$$84 > 25K$$

$$K < \frac{84}{25}$$

$$K < 3.6$$

$\therefore$ range of K is $0 < K < 3.6$

12. The block diagram of a feed back control system is shown in fig. Apply R-H criterion to determine the range of K for stability with $G(s) = \frac{K}{(s+4)(s+5)}$



$$1 - G(s) = 0$$

$$1 - \frac{G^2}{1-G^2} = 0$$

$$1 - 6^2 - 6^2 = 0$$

$$1 - 26^2 = 0$$

$$1 - 2 \frac{k^2}{(s+4)^2 (s+5)^2} = 0$$

$$(s+4)^2 (s+5)^2 - 2k^2 = 0$$

$$(s^2 + 16 + 8s)(s^2 + 25 + 10s) - 2k^2 = 0$$

$$s^4 + 85s^2 + 10s^3 + 16s^2 + 400 + 160s + 8s^3 + 200s + 80s^2 - 2k^2 = 0$$

$$s^4 + 18s^3 + 101s^2 + 360s + 400 - 2k^2 = 0$$

$$s^4 \quad 1 \quad 181 \quad (400 - 2k^2)$$

$$s^3 \quad 18 \quad 360 \quad 0$$

$$s^2 \quad 101 \quad (400 - 2k^2) \quad 0$$

$$s^1 \quad \frac{36360 - 18(400 - 2k^2)}{101} \quad 0 \quad 0$$

$$s^0 \quad 400 - 2k^2$$

$$\text{For row of } s^0, \quad 400 - 2k^2 > 0$$

$$400 > 2k^2$$

$$k^2 < \frac{400}{2}$$

$$k^2 < 200$$

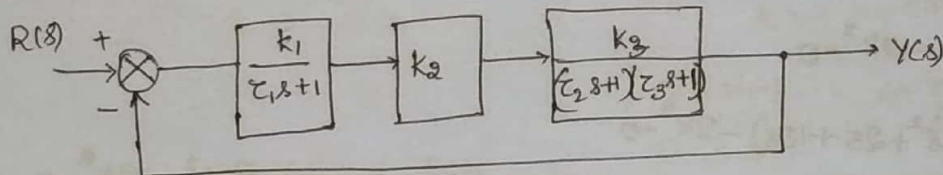
$$k < \sqrt{200}$$

$$k < 14.1421$$

For any value of k b/n $0 < k < \infty$, the term of s^1 is always positive. Thus range of k for stability is

$$0 < k < 14.1421$$

13. The control system for a wheelchair, designed for people paralysed below the neck, may be represented by the block diagram illustrated in fig. Typical values for the time constants are $T_1 = 0.5s$, $T_2 = 1s$ & $T_3 = 0.25s$. Determine the range of the gain, $K = K_1 K_2 K_3$ for a stable system.



Sol:- For the system given, $H(s) = 1$ &

$$\frac{G(s)}{R(s)} = \frac{K_1 K_2 K_3}{(\tau_1 s + 1)(\tau_2 s + 1)(\tau_3 s + 1)}$$

$$1 + \frac{K_1 K_2 K_3}{(\tau_1 s + 1)(\tau_2 s + 1)(\tau_3 s + 1)}$$

$$1 + G(s) \cdot H(s) = 1 + G(s) = 0$$

$$1 + \frac{K_1 K_2 K_3}{(\tau_1 s + 1)(\tau_2 s + 1)(\tau_3 s + 1)} = 0$$

$$(\tau_1 s + 1)(\tau_2 s + 1)(\tau_3 s + 1) + K_1 K_2 K_3 = 0$$

$$(\tau_1 \tau_2 s^2 + \tau_1 s + \tau_2 s + 1)(\tau_3 s + 1) + K_1 K_2 K_3 = 0$$

$$\tau_1 \tau_2 \tau_3 s^3 + \tau_1 \tau_2 s^2 + \tau_1 \tau_3 s^2 + \tau_1 s + \tau_2 \tau_3 s^2 + \tau_2 s + \tau_3 s + 1 + K_1 K_2 K_3 = 0$$

$$\tau_1 \tau_2 \tau_3 s^3 + (\tau_1 \tau_2 + \tau_2 \tau_3 + \tau_1 \tau_3) s^2 + (\tau_1 + \tau_2 + \tau_3) s + (1 + K) = 0$$

$$1 + \frac{K_1 K_2 K_3}{(1 + \tau_1 s)(1 + \tau_2 s)(1 + \tau_3 s)} = 0$$

$$(1 + \tau_1 s)(1 + \tau_2 s)(1 + \tau_3 s) + K_1 K_2 K_3 = 0$$

$$(1 + 0.5s)(1 + s)(1 + 0.25s) + K = 0$$

$$(1 + s + 0.5s + 0.5s^2)(1 + 0.25s) + K = 0$$

$$1 + 0.25s + s + 0.25s^2 + 0.5s + 0.125s^2 + 0.5s^2 + 0.125s^3 + K = 0$$

$$0.125s^3 + 0.875s^2 + 1.75s + (1 + K) = 0$$

s^3	0.125	1.75
s^2	0.875	$1+k$
s^1	$\frac{1.53125 - 0.125(1+k)}{0.875} = 0$	
s^0	$1+k$	

From row of s^0 , $1+k > 0$,
 $k > -1$

From row of s^1 , $1.53125 - 0.125(1+k) > 0$

$$1.53125 - 0.125 - 0.125k > 0$$

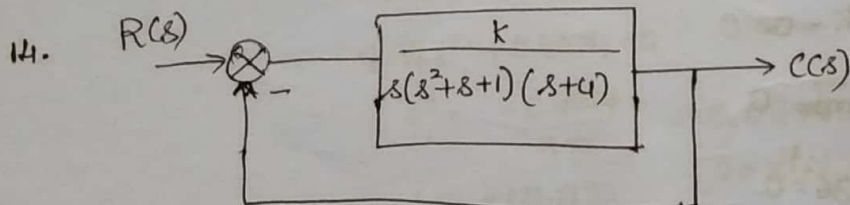
$$1.40625 - 0.125k > 0$$

$$1.40625 > 0.125k$$

$$k < 11.25$$

So, range of gain k for stability is,

$$-1 < k < 11.25$$



For the closed loop fed back system determine the range of k for which the system is stable & also the frequency of sustained oscillation.

Solⁿ:

$$1 + G(s)H(s) = 0$$

$$H(s) = 1$$

$$1 + \frac{k}{s(s^2 + s + 1)(s + 4)} = 0$$

$$s(s^2 + s + 1)(s + 4) + k = 0$$

$$(s^3 + s^2 + s)(s + 4) + k = 0$$

$$s^4 + 4s^3 + s^3 + 4s^2 + s^2 + 4s + k = 0$$

$$s^4 + 5s^3 + 5s^2 + 4s + k = 0$$

$$s^4 \quad 1 \quad 5 \quad K$$

$$s^3 \quad 5 \quad 4$$

$$s^2 \quad 21/5 \quad K$$

$$s^1 \quad \frac{21 \times 4 - 5K}{21/5} \quad 0$$

$$s^0 \quad K$$

$$\text{from } s^0, \quad K > 0.$$

$$\text{from } s^1, \quad \frac{21 \times 4 - 5K}{21/5} > 0$$

$$\frac{21 \times 4 - 5K}{5} > 0$$

$$21 \times 4 - 5K > 0$$

$$21 \times 4 > 5K$$

$$K < \frac{21 \times 4}{5}$$

$$K < 3.36.$$

$\therefore$ The range of K is

$$0 < K < 3.36.$$

In order to calculate the frequency of oscillation, 1st we need to get the value of K_{mar} . K_{mar} is the value of K for which the system is marginally stable system (row of zero element).

$$\therefore K_{mar} = 3.36.$$

$$A(s) = 21/5 s^2 + K = 0$$

$$\frac{21}{5} s^2 + K_{mar} = 0$$

$$\frac{21}{5} s^2 + 3.36 = 0$$

$$s^2 = -\frac{3.36}{21/5}$$

$$s^2 = -0.89$$

$$s = \pm j0.89$$

$$\text{WKT, } s = \pm j\omega$$

$$\therefore \boxed{\omega = 0.89 \text{ rad/sec}}$$

Repeated problem
(10)

15. For the system, $s^4 + 22s^3 + 10s^2 + 8s + K = 0$, find k_{max} & ω at k_{max}

Solⁿ:

s^4	1	10	K
s^3	22	1	0
s^2	9.95	K	0
s^1	$\frac{9.95 - 22K}{9.95}$	0	
s^0	K		

Marginal value of 'K' which makes row of s^1 as row of zeros,

$$9.95 - 22K_{max} = 0$$

$$K_{max} = \frac{9.95}{22}$$

$$K_{max} = 0.4524$$

$$A(s) = 9.95s^2 + K = 0$$

$$9.95s^2 + K_{max} = 0$$

$$9.95s^2 + 0.4524 = 0$$

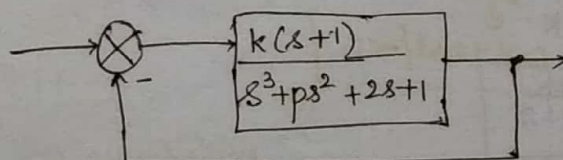
$$s^2 = \frac{-0.4524}{9.95} = -0.04546$$

$$s = \pm j0.2132$$

WKT, $s = \pm j\omega$,

$$\therefore \omega = 0.2132 \text{ rad/sec}$$

16) A given system oscillates with frequency 2 rad/sec.
(8.9) Find the values of 'K' & 'p'. No poles are in R.H.S.



March-02 Feb-03
Jan-08 July-08
June-10 Dec-11 Jan-14
July-15 (10m)

Solⁿ: As system oscillates, it is marginally stable & value of 'k' at this situation is k_{mar} .

As system is marginally stable, there must be row of zeros occurring in Routh's Array.

$$T.F = \frac{K(s+1)}{s^3 + ps^2 + 2s + 1}$$

$$1 + 1 \cdot \frac{K(s+1)}{s^3 + ps^2 + 2s + 1}$$

$$\boxed{d\tau = 0} \rightarrow \text{char eq}^n$$

$$\therefore 1 + \frac{K(s+1)}{s^3 + ps^2 + 2s + 1} = 0$$

$$s^3 + ps^2 + 2s + 1 + K(s+1) = 0$$

$$s^3 + ps^2 + 2s + 1 + Ks + K = 0$$

$$\therefore s^3 + ps^2 + (2+K)s + (1+K) = 0$$

$$\begin{array}{rcl} s^3 & 1 & (2+K) \\ s^2 & p & (1+K) \\ s^1 & \frac{(2+K)p - (1+K)}{p} & 0 \\ s^0 & 1+K & \end{array}$$

At marginal value of k is k_{mar} ,

$$(2+K)p - (1+K) = 0$$

$$(2+K)p = 1+K$$

$$\boxed{p = \frac{1+K}{K+2}}$$

Now at this value,

$$A(s) = ps^2 + (1+K) = 0$$

$$s^2 = -\frac{(1+K)}{p}$$

$$s = \pm j\sqrt{\frac{1+K}{p}}$$

Compare with $s = \pm j\omega$ & freq ω is 2.

$$\therefore \sqrt{\frac{1+K}{P}} = \omega = 2.$$

$$\left(\sqrt{\frac{1+K}{P}}\right)^2 = (2)^2$$

$$\frac{1+K}{P} = 4$$

$$P = \frac{1+K}{4}$$

$$\frac{1+K}{2+K} = \frac{1+K}{4}$$

$$2+K = 4 \Rightarrow K = 4-2 = 2.$$

$$\therefore \boxed{K_{mar} = 2}$$

$$\therefore p = \frac{K_{mar} + 1}{4} = \frac{2+1}{4} = \frac{3}{4} = 0.75$$

$$\therefore \boxed{p = 0.75}$$

17. The open loop transfer funⁿ of a feedback system is,
(8.11)

$$G(s).H(s) = \frac{k(s+5)}{s(1+Ts)(1+2s)}$$

Parameters k & T are represented on a plane with k on x -axis & T on y -axis. Determine region in which a closed loop system is stable.

Solⁿ: The char eqⁿ of the sys is,

$$1 + G(s).H(s) = 0$$

$$1 + \frac{k(s+5)}{s(1+Ts)(1+2s)}$$

$$s(1+Ts)(1+2s) + k(s+5) = 0$$

$$(s+Ts^2)(1+2s) + ks + k5 = 0$$

$$s + 2s^2 + Ts^2 + 2Ts^3 + ks + k5 = 0$$

$$2Ts^3 + (2+T)s^2 + (k+1)s + 5k = 0$$

Routh Array is,

$$\begin{array}{rcl}
 s^3 & 2T & (k+1) \\
 s^2 & (T+2) & 5k \\
 s^1 & \frac{(T+2)(k+1) - 10Tk}{(T+2)} & 0 \\
 s^0 & 5k & 0
 \end{array}$$

$$s^0, \quad 5k > 0 \quad \text{or} \quad k > 0$$

$$\begin{aligned}
 s^1, \quad & (T+2)(k+1) - 10Tk > 0 \\
 & kT + T + 2k + 2 - 10Tk > 0 \\
 & -9Tk + T + 2k + 2 > 0 \\
 & k(2-9T) + T + 2 > 0
 \end{aligned}$$

$$k \geq \frac{-(T+2)}{2-9T}$$

$$\text{ie } \boxed{k < \frac{T+2}{9T-2}}$$

$$\therefore \boxed{0 < k < \frac{T+2}{9T-2}}$$

$\therefore$ Region in which closed loop system is stable.

18. Using R-H criteria determine stability of following

(8.12)

1) Its loop transfer f_n has poles at $s=0, s=-1, s=-3$ & zero at $s=-5$, gain k of forward path is 10.

Solⁿ: Loop T.F has pole at $s=0, s=-1, s=-3$ & zeroes at $s=-5$ & gain $k=10$.

$$G(s) \cdot H(s) = \frac{10(s+5)}{s(s+1)(s+3)}$$

$\therefore$ Char eqⁿ $1 + G(s)H(s) = 0$

$$1 + \frac{10(s+5)}{s(s+1)(s+3)} = 0$$

$$s(s^2+4s+3) + 10s+5 = 0$$

$$s^3 + 4s^2 + 13s + 5 = 0$$

$$s^3 \quad 1 \quad 13$$

$$s^2 \quad 4 \quad 50$$

$$s^1 \quad 0.5 \quad 0$$

$$s^0 \quad 50$$

Since no sign change in the 1st column hence sys is absolutely stable in nature.

19. Determine the ranges of k such that the Char eqⁿ

$$s^3 + 3(k+1)s^2 + (7k+5)s + (4k+7) = 0 \text{ has roots more -ve than}$$

$$s = -1.$$

Aug-01, Feb-03, Aug-03
July-07, 18 Jan 11
(12m)

Solⁿ: put $s = z-1$,

$$s^3 + 3(k+1)s^2 + (7k+5)s + (4k+7) = 0$$

$$(z-1)^3 + (3k+3)(z-1)^2 + (7k+5)(z-1) + 4k+7 = 0$$

$$(z^2+1-2z)(z-1) + (3k+3)(z^2+1-2z) + 7kz+5z-7k-5+4k+7 = 0$$

$$z^3 + z - 2z^2 - z^2 - 1 + 2z + 3kz^2 + 3k - 6kz + 3z^2 + 3 - 6z + 7kz + 5z - 3k + 2 = 0$$

$$z^3 + 3kz^2 + 2z + kz + 4 = 0$$

$$\boxed{z^3 + 3kz^2 + (2+k)z + 4 = 0}$$

Routh's Array is,

$$\begin{array}{ccc}
 z^3 & 1 & (k+2) \\
 z^2 & 3k & 4 \\
 z^1 & \frac{3k(k+2)-4}{3k} & 0 \\
 z^0 & 4 &
 \end{array}$$

From z^2 , $3k > 0$
 $\boxed{k > 0}$

$$3k(k+2)-4 > 0$$

$$3k^2 + 6k > 4 \quad 3k^2 + 6k - 4 > 0$$

$$3k^2 > 4 - 6k \quad (\div 3) \quad k^2 + 2k - 1.33 > 0$$

$$k^2 > \frac{4-6k}{3} \quad (k-0.5275)(k+2.5275) > 0$$

$$k > 0.5275 \text{ \& } k < -2.5275$$

$\therefore$ Range of k is $\boxed{0.5275 < k < \infty}$, to have roots more
 -ve than $s=-1$ is located to the left of $s=-1$.

20. Determine the location of roots with respect to $s=-2$,
 given that the char eqⁿ is,

$$s^4 + 10s^3 + 36s^2 + 10s + 15 = 0$$

Solⁿ: Shift the origin w.r.t $s=-2$

$$\therefore \text{ put } s = z - 2$$

$$(z-2)^4 + 10(z-2)^3 + 36(z-2)^2 + 10(z-2) + 15 = 0$$

$$z^4 + 2z^3 + 0(z^2) + 14z + 15 = 0$$

RH array is,

z^4	1	0	15
z^3	+2	14	0
z^2	-1	15	0
z^1	18.28	0	
z^0	15		

There are 2 sign changes in the 1st column hence there are 2 roots to the RHS of $s = -2$ & remaining 2 are left of $s = -2$.

Q1. The close loop T.F of a position control system is given by, $\frac{C(s)}{R(s)} = \frac{K}{s^4 + 6s^3 + 30s^2 + 60s + K}$

- 1) Determine the range of k must lie for the system to be stable.
- 2) What should be the upper limit on k if all the closed loop poles are required to be to the left of the line $\sigma = -1$.

Solⁿ: $s^4 + 6s^3 + 30s^2 + 60s + k = 0$

Using Routh's array,

s^4	1	30	k
s^3	6	60	0
s^2	20	k	
s^1	$\frac{1200-6k}{20}$	0	
s^0	k		

$k > 0$

$\frac{1200-6k}{20} > 0$

$1200-6k > 0$

$6k < 1200$

$k < 200$

$\therefore \boxed{0 < k < 200}$

Replace s by $(z-1)$

$$(z-1)^4 + 6(z-1)^3 + 30(z-1)^2 + 60(z-1) + k = 0$$

$$(z-1)^2 (z-1)^2$$

$$z^4 - 4z^3 + 6z^2 - 4z + 1$$

$$+ 6z^3 - 18z^2 + 18z - 6$$

$$30z^2 - 60z + 30$$

$$+ 60z - 60 + k$$

$$z^4 + 2z^3 + 18z^2 + 14z - 35 + k = 0$$

z^4	1	18	$(-35+k)$
z^3	2	14	0
z^2	11	$(-35+k)$	0
z^1	$\frac{224-2k}{11}$	0	
z^0	$-35+k$		

$$-35+k > 0$$

$$\boxed{k > 35} \rightarrow \text{lower limit}$$

$$\frac{224-2k}{11} > 0$$

$$224 - 2k > 0$$

$$224 < 2k$$

$$\boxed{k < 112} \rightarrow \text{upper limit}$$

22) An open loop T-F has poles at $s=0$, $s=-1$ & a zero at $s=-2$. Check the range of k for the stability of the closed loop system.

$$O.L.T.F = \frac{k(s+2)}{s(s+1)} = G(s)H(s)$$

$$1 + G(s)H(s) = 1 + \frac{k(s+2)}{s(s+1)} = 0$$

$$s^2 + s(k+1) + 2k = 0$$

Routh's Array,

$$s^2 \quad 1 \quad 2k$$

$$s^1 \quad (k+1) \quad 0$$

$$s^0 \quad 2k$$

$$s^0 \rightarrow 2k > 0 \text{ i.e. } k > 0$$

$$s^1 \rightarrow 1+k > 0 \text{ i.e. } k > -1$$

$\therefore \boxed{0 < k < \infty}$ the closed loop system is stable.

23. $s^6 + 4s^5 + 3s^4 - 16s^2 - 64s - 48 = 0$. Find the number of roots of this equation with positive real part, zero real part & negative real

Sol:

s^6	1	3	-16	-48
s^5	4	0	-64	0
s^4	3	0	-48	
s^3	(0)12	0	0	
s^2	0[E]	-48	0	
s^1	$\frac{576}{e}$	0	0	
s^0	-48			

Jan-14
July-14
(8m)

$$A(s) = 3s^4 - 48 = 0$$

$$\frac{dA}{ds} = 12s^3$$

$$\lim_{e \rightarrow 0} \frac{576}{e} = +\infty$$

$\therefore$ One sign change & system is unstable. Thus there is one root in R.H.S of s-plane i.e. with +ve real part.

$$A(s) = 3s^4 - 48 = 0$$

$$s^2 = y \quad -3y^2 = 48$$

$$y^2 = +48/3 = 16$$

$$y = \pm \sqrt{16}$$

$$\boxed{y = \pm 4}$$

$$s^2 = +4$$

$$\boxed{s = \pm 2}$$

$$s^2 = -4$$

$$\boxed{s = \pm j2}$$

So, $s = \pm j2$ are the 2 roots on imaginary axis i.e. with zero real part. Root in RHS indicated by a sign change is $s = +2$ as obtained by solving $A(s) = 0$.

Total roots = 6 roots $n=6$

Roots with positive real part = 1

Roots with zero real part = 2

Roots with negative real part = $6 - 2 - 1 = 3$

24. For unity feedback system, system is marginally stable & oscillates with frequency 4 rad/sec. Find K_{max} & q .

$$G(s) = \frac{4}{(s^2 + qs + 2k)s}$$

Solⁿ: Characteristic eqⁿ: $1 + G(s) = 0$ as $H(s) = 1$

$$1 + \frac{4}{(s^2 + qs + 2k)s} = 0$$

$$s^3 + qs^2 + 2ks + 4 = 0$$

s^3	1	$2k$
s^2	q	4
s^1	$\frac{2kq - 4}{q}$	
s^0	4	

For marginal value of k ,

$$2kq - 4 = 0$$

$$2kq = 4$$

$$q = \frac{4}{2k}$$

Now for this value of k ,

$$A(s) = qs^2 + 4 = 0$$

$$s^2 = -4/q$$

$$s = \pm j \frac{2}{\sqrt{q}}$$

$$\omega = \text{Frequency} = 4$$

$$\omega = (4)^2 = \left(\frac{2}{\sqrt{q}}\right)^2$$

$$\frac{4}{q} = 16$$

$$q = \frac{4}{16} = 0.25$$

$$q = 0.25$$

$$q = 4/2k$$

$$k = \frac{4}{2q} = \frac{4}{2 \times 0.25} = 8$$

$$K_{max} = 8 \quad q = 0.25$$

25. Find the range of k for which the system, whose char eqⁿ is given below, is stable.

July 16
(8m)

$$F(s) = s^3 + (k+0.5)s^2 + 4ks + 50 = 0.$$

Solⁿ:

s^3	1	$4k$
s^2	$k+0.5$	
s^1	$\frac{(k+0.5)(4k) - 50}{(k+0.5)}$	
s^0	50	

$$k+0.5 > 0$$

$$\boxed{k > -0.5}$$

$$(k+0.5)(4k) - 50 > 0$$

$$4k^2 + 2k - 50 > 0$$

$$\therefore k = \frac{-1 \pm \sqrt{1 + 25 \times 4 \times 2}}{2 \times 2}$$

The two values of the roots are
3.2943, -3.7943, hence we can write,

$$= (k - 3.2943)(k + 3.7943) > 0$$

$$\boxed{k > 3.2943 \text{ \& } k > -3.7943}$$

$\therefore$ Range of values of k for the system stability is,

$$\boxed{3.2943 < k < \infty}$$

26. Examine stability of $F(s) = s^3 + 6s^2 + 11s + 6 = 0$

Solⁿ:

s^3	1	11
s^2	6	6
s^1	$b_1 = 10$	0
s^0	6	

$$b_1 = \frac{6 \times 11 - 1 \times 6}{6} = \frac{66 - 6}{6} = \frac{60}{6} = 10$$

As there is no sign change in first column, system is stable.

Comment on the stability of a system using Routh's stability criteria whose characteristic equation is:

$$s^4 + 2s^3 + 4s^2 + 6s + 8 = 0$$

How many poles of systems lie in right half of s -plane

July -11
(4m)

Solⁿ:

$$s^4 + 2s^3 + 4s^2 + 6s + 8 = 0$$

$$s^4 \quad 1 \quad 4 \quad 8$$

$$s^3 \quad 2 \quad 6 \quad 0$$

$$s^2 \quad +1 \quad 8 \quad 0$$

$$\textcircled{1} \left(\begin{array}{l} s^1 \\ s^0 \end{array} \right) \begin{array}{l} -10 \\ +8 \end{array}$$

$$\textcircled{2} \left(\begin{array}{l} s^1 \\ s^0 \end{array} \right) \begin{array}{l} -10 \\ +8 \end{array}$$

There are 2 sign changes in the first column hence system is unstable with 2 poles in right half of s-plane.

27. A unity feedback control system has:

$$G(s) = \frac{20K}{s[(s+1)(s+5)+20]} \quad \text{where } r(t) = 2t$$

Jan 11
(8m)

It is desired that for ramp input $e_{ss} \leq 1.5$. What minimum value must K have for this condition to be satisfied? With this K , is the system stable?

Solⁿ: For ramp input, $r(t)$, $A=2$

$$K_v = \lim_{s \rightarrow 0} s \cdot G(s)H(s) = \lim_{s \rightarrow 0} s \cdot \frac{20K}{s[(s+1)(s+5)+20]} = \frac{20K}{25}$$

$$e_{ss} = \frac{A}{K_v} = \frac{2}{\frac{20K}{25}} = \frac{2 \times 25}{20K} = 1.5 \Rightarrow K = \frac{2 \times 25}{20 \times 1.5} = 1.667 \text{ min}$$

$$\boxed{K = 1.667 \text{ (min)}}$$

$$\text{For } K = 1.667 \quad G(s)H(s) = \frac{20K}{s[(s+1)(s+5)+20]}$$

$$\text{Char eqⁿ: } 1 + G(s)H(s) = 0$$

$$1 + \frac{20(1.667)}{s[s^2 + 5s + 8 + 20]} = 0$$

$$\frac{s[s^2 + 6s + 25] + 33.33}{s[s^2 + 6s + 25]} = 0$$

$$\boxed{s^3 + 6s^2 + 25s + 33.33 = 0}$$

$$s^3 + 6s^2 + 25s + 33.33 = 0$$

$$s^3 \quad 1 \quad 25$$

$$s^2 \quad 6 \quad 33.33$$

$$s^1 \quad 19.44 \quad 0$$

$$s^0 \quad 33.33$$

As there is no sign change in the 1st column, system is stable for $k = 1.667$.

28. Comment on stability using Routh criteria of Characteristic equation is:

$$Q(s) = s^5 + 2s^4 + 3s^3 + 4s^2 + 5s + 6 = 0$$

How many poles lie in right half of s-plane?

Sol:

$$s^5 \quad 1 \quad 3 \quad 5$$

$$s^4 \quad 2 \quad 4 \quad 6$$

$$s^3 \quad 1 \quad 2 \quad 0$$

$$s^2 \quad 0+E \quad 6$$

$$s^1 \quad \frac{2E-6}{E} \quad 0$$

$$s^0 \quad 6$$

$$\lim_{E \rightarrow 0} \frac{2E-6}{E} \rightarrow -ve$$

Thus there are 2 sign changes in the 1st column hence 2 poles lie in Right half of s-plane.

$\therefore$ System is unstable.

29. Explain briefly the Routh-Hurwitz criterion & use it to determine the roots i) in RHS ii) LHS & iii) On jw axis of s-plane for the polynomial.

$$P(s) = s^6 + s^5 + 5s^4 + s^3 + 2s^2 - 2s - 8$$

March 02
(10m)

Solⁿ: $P(s) = s^6 + s^5 + 5s^4 + s^3 + 2s^2 - 2s - 8$

The Routh's array is:

$$s^6 \quad 1 \quad 5 \quad 2 \quad -8$$

$$s^5 \quad 1 \quad 1 \quad -2 \quad 0$$

$$s^4 \quad 4 \quad 4 \quad -8 \quad 0$$

$$s^3 \quad 0 \quad 0 \quad 0 \quad \leftarrow \text{special case 2}$$

$$s^2 \quad 2 \quad -8 \quad 0$$

$$s^1 \quad -2 \quad 0$$

$$s^0 \quad -8$$

$$A(s) = 4s^4 + 4s^2 - 8 = 0$$

$$\frac{dA(s)}{ds} = 16s^3 + 8s = 0$$

$$s^2 = -2, +1$$

$$s = \pm j\sqrt{2}, s = \pm 1$$

There is one sign change so
one root in RHS of s-plane.

Order of eqⁿ = 6

$\therefore$ Out of 6 roots

in RHS = 1 root

in jw axis = 2 roots ($\pm j\sqrt{2}$)

in LHS = 3 roots.

30. For the characteristic equations given below
determine the number of roots with positive
real part.

Aug-06

(10m).

i) $s^6 + s^5 + 3s^4 + 2s^3 + 5s^2 + 3s + 1 = 0$

ii) $s^8 + s^7 + 4s^6 + 3s^5 + 14s^4 + 11s^3 + 20s^2 + 9s + 9 = 0$

i) $s^6 \quad 1 \quad 3 \quad 5 \quad 1$

$$s^5 \quad 1 \quad 2 \quad 3 \quad 0$$

$$s^4 \quad 1 \quad 2 \quad 1 \quad 0$$

$$s^3 \quad 0 \quad 2 \quad 0 \quad \leftarrow \text{special case 1}$$

$$s^2$$

$$s^1$$

$$s^0$$

31. Using R-H criterion determine the stability of the system having the characteristic eqⁿ:

$$s^6 + 2s^5 + 5s^4 + 8s^3 + 8s^2 + 8s + 4 = 0$$

Jan 18
Jan 15 (6m)

Solⁿ:

$$s^6 \quad 1 \quad 5 \quad 8 \quad 4$$

$$s^5 \quad 2 \quad 8 \quad 8 \quad 0$$

$$s^4 \quad 1 \quad 4 \quad 4 \quad 0$$

$$s^3 \quad 0 \quad 4 \quad 8 \quad 0$$

$$s^2 \quad 2 \quad 4$$

$$s^1 \quad 0 \quad 4 \quad 0 \quad \leftarrow \text{special case 2}$$

$$s^0 \quad 4$$

← Special case 2

$$A(s) = s^4 + 4s^2 + 4$$

$$\frac{dA(s)}{ds} = 4s^3 + 8s$$

$$A'(s) = 2s^2 + 4$$

$$\frac{dA'(s)}{ds} = 4s$$

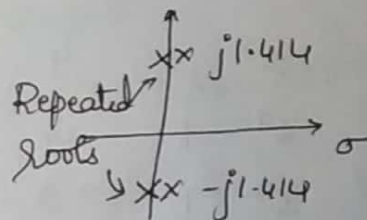
No sign change in 1st column

so no roots in RHS of s-plane. jw

$$\text{Solve: } A(s) = s^4 + 4s^2 + 4 = 0$$

$$s^2 = -2, \quad s^2 = -2$$

$$\therefore s = \pm j1.414, \quad s = \pm j1.414$$



There are repeated roots on jw axis
hence system is unstable in nature.

32. The open loop transfer function of a unity feedback system is given by,

$$G(s) = \frac{k}{s(1+sT_1)(1+sT_2)}$$

Derive an expression for gain k in terms of T_1 & T_2 for the stability of the system.

April 96, (6m).

$$s^6 + s^5 + 3s^4 + 2s^3 + 5s^2 + 3s + 1 = 0$$

$$s \rightarrow \frac{1}{z}$$

$$\frac{1}{z^6} + \frac{1}{z^5} + \frac{3}{z^4} + \frac{2}{z^3} + \frac{5}{z^2} + \frac{3}{z} + 1 = 0$$

$$\times z^6$$

$$1 + z + 3z^2 + 2z^3 + 5z^4 + 3z^5 + z^6 = 0$$

$$z^6 + 3z^5 + 5z^4 + 2z^3 + 3z^2 + z + 1 = 0$$

z^6	1	5	3	1
z^5	3	2	1	0
z^4	4.33	2.67	1	
z^3	0.150	0.307	0	
z^2	-6.192	1		
z^1	0.33	0		
z^0				

Thus there are 2 sign changes in 1st column of Routh's array

$\therefore$ 2 roots in R.H.S.

$$ii) s^8 + s^7 + 4s^6 + 3s^5 + 14s^4 + 11s^3 + 20s^2 + 9s + 9 = 0$$

s^8	1	4	14	20	9
s^7	1	3	11	9	0
s^6	1	3	11	9	0
s^5	0.6	0.12	0.22	0	
s^4	1	7.33	9	0	
s^3	-3.2	-2.3	0		
s^2	6.6145	9	0		
s^1	13.2839	0			
s^0	9				

← special case 2.

$$A(s) = s^6 + 3s^4 + 11s^2 + 9 = 0$$

$$\frac{dA(s)}{ds} = 6s^5 + 44s^3 + 22s = 0$$

$\therefore$ There are 2 sign changes in 1st column of Routh's array.

$\therefore$ 2 roots in R.H.S.

$\therefore$ System is unstable.

$$P(s) = (s')^4 - 2(s')^3 + 3(s')^2 - 3s' + 2$$

Routh's array is,

$$\begin{array}{r|rrrr} s'^4 & 1 & 3 & 2 & \\ \textcircled{1} & & & & \\ (s')^3 & -2 & -3 & 0 & \\ \textcircled{2} & & & & \\ (s')^2 & 1.5 & 2 & & \\ \textcircled{3} & & & & \\ (s')^1 & -0.33 & 0 & & \\ \textcircled{4} & & & & \\ (s')^0 & 2 & & & \end{array}$$

There are 4 sign changes
hence 4 roots are at
right of $s = -1$

Now checking for no. of roots RHS of $s = -1/2$
replace $s = (z - 0.5)$

$$\begin{aligned} P(s) &= (z-0.5)^4 - 2(z-0.5)^3 + 3(z-0.5)^2 - 3(z-0.5) + 2 \\ &= (z-0.5)^2(z-0.5)^2 + 2[z^3 - (0.5)^3 - 3(z)(0.5)[z-0.5]] + 3[z^2 + (0.5)^2 - 2z(0.5)] \\ &\quad + z + 0.5 \\ &= (z^2 + (0.5)^2 - z)(z^2 + (0.5)^2 - z) + 2[z^3 - (0.5)^3 - 3z^2(0.5) + 3z(0.5)(0.5)] \\ &\quad + 3z^2 + 3(0.5)^2 - z + z + 0.5 \\ &= z^4 + \underline{z^2 \cdot 0.25} - \underline{z^3} + \underline{0.25z^2} + (0.25)(0.25) - \underline{0.25z} - \underline{z^3} - \underline{0.25z} + \underline{z^2} \\ &\quad + \underline{2z^3} - \underline{2(0.5)^3} - \underline{3z^2} + \underline{3z(0.5)} + \underline{3z^2} + \underline{3(0.25)} + 0.5 \end{aligned}$$

$$P(s) = z^4 + 1.5z^2 + z + 1.0625$$

Solⁿ: The characteristic eqⁿ is:

$$1 + G(s)H(s) = 0$$

$$1 + \frac{K}{s(1+sT_1)(1+sT_2)} = 0$$

$$T_1 T_2 s^3 + s^2(T_1 + T_2) + s + K = 0$$

Routh's array is:

$$s^3 \quad T_1 T_2 \quad 1$$

$$s^2 \quad (T_1 + T_2) \quad K$$

$$s^1 \quad \frac{(T_1 + T_2) - K T_1 T_2}{(T_1 + T_2)} \quad 0$$

$$s^0 \quad K$$

$$\text{From } s^0, \boxed{K > 0}$$

$$\text{From } s^1,$$

$$\frac{(T_1 + T_2) - K T_1 T_2}{(T_1 + T_2)} > 0$$

$$T_1 + T_2 > K T_1 T_2$$

$$\frac{T_1 + T_2}{T_1 T_2} > K$$

$$K < \frac{T_1 + T_2}{T_1 T_2} = \frac{T_1}{T_1 T_2} + \frac{T_2}{T_1 T_2}$$

$$K < \frac{1}{T_2} + \frac{1}{T_1}$$

So $\boxed{0 < K < \frac{1}{T_1} + \frac{1}{T_2}}$ is the range of K for stability.

33. The polynomial $P(s) = s^4 + 2s^3 + 3s^2 + s + 1$ has all its roots in LHS of a plane. Use R-H Criterion to determine the no. of roots of $P(s)$ lying between $s = -\frac{1}{2}$ & $s = -1$.

Aug-05
(10m)

$$P(s) = s^4 + 2s^3 + 3s^2 + s + 1$$

substitute $s = s' - 1$

$$P'(s) = (s' - 1)^4 + 2(s' - 1)^3 + 3(s' - 1)^2 + (s' - 1) + 1$$

$$= (s' - 1)^2 (s' - 1)^2 + 2[s'^3 - 3s'^2 + 3s' - 1] + 3[s'^2 - 2s' + 1] + s'$$

$$= [s'^2 - 2s'] [s'^2 - 2s'] + 2s'^3 - 6s'^2 + 6s' - 2 + 3s'^2 - 6s' + 3 + s'$$

$$= s'^4 + s'^2 - 2s'^3 + s'^2 + 1 - 2s' - 2s'^3 - 2s' + 4s'^2 + 2s'^3 + 3s'^2 - 6s' + 1$$

Routh array is

$$\begin{array}{cccc} z^4 & 1 & 1.5 & 1.0625 \\ z^3 & 0 & 1 & 0 \leftarrow \text{special case 1.} \end{array}$$

$$z^2$$

z^1 $\therefore$ Rewrite the eqⁿ

$$P(s) = z^4 + 1.5z^2 + z + 1.0625$$

$$z^0$$

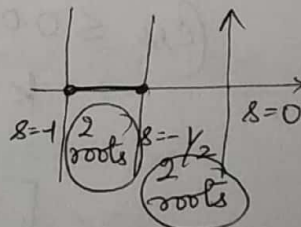
$$z \rightarrow \frac{1}{z} = \left(\frac{1}{z}\right)^4 + 1.5\left(\frac{1}{z}\right)^2 + \frac{1}{z} + 1.0625$$

$$P(s) = z^4 1.0625 + z^3 + 1.5z^2 + 1$$

$$\begin{array}{cccc} z^4 & 1.0625 & 1.5 & 1 \\ z^3 & 1 & 0 & 0 \\ z^2 & \textcircled{1} \swarrow 1.5 & 1 & \\ z^1 & \textcircled{1} \swarrow 0.666 & 0 & \\ z^0 & \textcircled{2} \swarrow 1 & & \end{array}$$

Since there are 2 sign change 2 roots lie on
RHS of $s = -\frac{1}{2}$

Thus there are 2 roots lying
b/n $s = -\frac{1}{2}$ & $s = -1$



34. The open loop transfer function of a unity feedback system is:

$$G(s) = \frac{K(s+2)}{s(s+3)(s^2+5s+10)}$$

i) Find the value of K so that the steady state error for the i/p $x(t) = t \cdot u(t)$ is less than or equal to 0.01.

ii) For the value of K found in part i), Verify whether the closed loop system is stable or not using R-H criterion

Jan 14
July-05, 17, 19
(6m).

Solⁿ:

i) $x(t) = t \cdot u(t)$ is ramp i/p with $A=1$.

$$K_V = \lim_{s \rightarrow 0} s \cdot G(s) \cdot H(s) = \lim_{s \rightarrow 0} s \cdot \frac{K(s+2)}{s(s+3)(s^2+5s+10)} = \frac{2K}{30}$$

$$\boxed{K_V = 0.0666K}$$

$$e_{ss} = \frac{A}{K_V} = \frac{1}{0.0666K} = \frac{15}{K} = 0.01$$

($e_{ss} \leq 0.01$) given

$$\therefore K = \frac{15}{0.01} = 1500$$

$$\boxed{K = 1500}$$

ii) $K = 1500$

$$1 + G(s)H(s) = 1 + \frac{1500(s+2)}{s(s+3)(s^2+5s+10)} = 0$$

$$(s^2+3s)(s^2+5s+10) + 1500s + 3000 = 0$$

$$s^4 + 5s^3 + 10s^2 + 3s^3 + 15s^2 + 30s + 1500s + 3000 = 0$$

$$s^4 + 8s^3 + 25s^2 + 1530s + 3000 = 0$$

$$\begin{array}{rcccc}
 s^4 & 1 & 25 & 3000 & \\
 s^3 & 8 & 1530 & 0 & \\
 s^2 & -166.25 & 3000 & 0 & \\
 s^1 & -1674.36 & 0 & & \\
 s^0 & 3000 & & &
 \end{array}$$

There are 2 sign changes in the 1st column hence the system is unstable in nature.

35. Find the range of K for the system to be stable using R-H criterion.

June 13
(6m)

$$G(s)H(s) = \frac{K(1-s)}{s(s^2+5s+9)}$$

Solⁿ:

Characteristic eqⁿ is: $1+G(s)H(s) = 0$

$$1 + \frac{K(1-s)}{s(s^2+5s+9)} = 0$$

$$s^3 + 5s^2 + 9s + K - sK = 0$$

$$s^3 + 5s^2 + (9-K)s + K = 0$$

Routh array is:

$$\begin{array}{rcc}
 s^3 & 1 & (9-K) \\
 s^2 & 5 & K \\
 s^1 & \frac{45-5K-K}{5} & 0 \\
 s^0 & K &
 \end{array}$$

From row of s^0 , $K > 0$

From row of s^1 , $\frac{45-6K}{5} > 0$

$$45 - 6K > 0$$

$$45 > 6K$$

$$\frac{45}{6} > K$$

$$K < \frac{45}{6}$$

$$K < 7.5$$

Hence the range of K for stability is

$$0 < K < 7.5$$

36) The characteristics equation for certain feedback control system are given below. Determine the system is stable or not & find the value of k for a stable system: $s^3 + 3ks^2 + (k+2)s + 4 = 0$

July 14
(6m)

Solⁿ:

s^3	1	$k+2$
s^2	$3k$	4
s^1	$\frac{3k^2 + 6k - 4}{3k}$	0
s^0	4	

from s^2 $3k > 0$
 $\boxed{k > 0}$

from s^1 $\frac{3k^2 + 6k - 4}{3k} > 0$

Hence the range of positive values of k for stability is $\boxed{0 < k < 0.5275}$

$(k - 0.5275)(k + 2.5275) > 0$
 $\therefore k > 0.5275$ &
 $k > -2.5275$
 or $k < 0.5275$ &
 $k < -2.5275$

37) By applying Routh Criteria, discuss the stability of the closed loop system as function of k for the following Open loop transfer function: $G(s) = \frac{k(s+1)}{s(s-1)(s^2+4s+16)}$

Solⁿ: Characteristic eqⁿ is

$$1 + G(s)H(s) = 0$$

$$1 + \frac{k(s+1)}{s(s-1)(s^2+4s+16)} = 0$$

$$(s^2 - s)(s^2 + 4s + 16) + k(s+1) = 0$$

$$s^4 + 4s^3 + 16s^2 - s^3 - 4s^2 - 16s + ks + k = 0$$

July 14
Jan 18 (6m)

$$s^4 + 8s^3 + 12s^2 + s(k-16) + k = 0$$

Routh array is:

$$\begin{array}{cccc} s^4 & 1 & 12 & k \\ s^3 & 8 & (k-16) & 0 \\ s^2 & \frac{52-k}{3} & k & 0 \\ s^1 & \frac{52-k(k-16)-3k}{3} & 0 & \\ s^0 & k & & \end{array}$$

from s^0 , $k > 0$

from s^1 ,

$$\frac{52-k}{3}(k-16) - 3k > 0$$

$$(52-k)(k-16) - 9k > 0$$

$$52k - 832 - k^2 + 16k - 9k > 0$$

$$59k - k^2 - 832 > 0$$

$$k^2 - 59k + 832 < 0$$

$$(k - 23.31)(k - 35.684) < 0$$

$$k < 23.31 \text{ \& } k > 35.684$$

$$\text{or } k > 23.31 \text{ \& } k < 35.684$$

Thus $k > 23.31$ but

$k < 35.684$ is the ultimate range of k for stability.

$$\boxed{23.31 < k < 35.684}$$

38) What value of k makes the following unity feedback system stable?

$$G(s) = \frac{k(s+1)^2}{s^3}$$

Jan 14
(4m)

Sol: Characteristic eqⁿ is: $1 + G(s)H(s) = 0$

$$1 + \frac{k(s+1)^2}{s^3} = 0$$

$$s^3 + k(s^2 + 1 + 2s) = 0$$

$$s^3 + ks^2 + k + 2ks = 0$$

$$s^3 + s^2 + 2ks + k = 0$$

Routh's array is:

$$s^3 \quad 1 \quad 2K$$

$$s^2 \quad K \quad K$$

$$s^1 \quad \frac{2K^2 - K}{K} \quad 0$$

$$s^0 \quad K$$

From s^0 & s^2 $K > 0$

From s^1 $\frac{2K^2 - K}{K} > 0$

$$K(2K - 1) > 0$$

$$2K - 1 > 0$$

$$2K > 1$$

$$K > \frac{1}{2}$$

$\therefore$ Range of K for stability is

$$\boxed{\frac{1}{2} < K < \infty}$$

39) The open loop TF of a unity feedback control system is given by $G(s) = \frac{K}{(s+2)(s+4)(s^2+6s+25)}$

Determine the range of K for system stability. What is the value of K which gives sustained oscillations? What is the oscillation frequency?

May 10
(10m)

Solⁿ:

$$1 + G(s)H(s) = 0$$

$$1 + \frac{K}{(s+2)(s+4)(s^2+6s+25)} = 0$$

$$\therefore s^4 + 12s^3 + 69s^2 + 198s + (200 + K) = 0$$

$$s^4 \quad 1 \quad 69 \quad 198 \quad 200 + K$$

$$s^3 \quad 12 \quad 198 \quad 0$$

$$s^2 \quad 52.5 \quad (200 + K) \quad 0$$

$$s^1 \quad \frac{7995 - 12K}{52.5} \quad 0$$

$$s^0 \quad 200 + K$$

From s^0 :

$$200 + K > 0$$

$$\boxed{K > -200}$$

From s^1 :

$$7995 - 12K > 0$$

$$K < \frac{7995}{12} < 666.25$$

$$\therefore \boxed{-200 < K < 666.25}$$

Find k such that it makes row of s^1 as row of zeros.

For marginal value of k

$$7995 - 12k = 0$$

$$\boxed{k_{\text{mar}} = 666.25}$$

$$A(s) = 52.5s^2 + (200 + k) = 0$$

$$s^2 = \frac{-(200 + k)}{52.5}$$

$$\text{At } k_{\text{mar}}, s^2 = -\frac{(200 + 666.25)}{52.5}$$

$$\boxed{s = \pm j 4.062}$$

$$s = \pm j\omega$$

$$\text{Oscillation freq. : } \boxed{\omega = 4.062 \text{ rad/sec.}}$$